

Important Characteristics, Weaknesses and Errors in German Equity Data from Thomson Reuters Datastream and their Implications for the Size Effect

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We examine the characteristics and the quality of equity data from Thomson Reuters Datastream (Datastream) to evaluate, whether Datastream can be used as the primary data source for academic studies on the German stock market. We document market coverage issues in Datastream before 1990. Additionally, Datastream's coverage of dividends is sufficient for standard academic use only from 1990 onwards. After 1990 systematic data errors are rare in Datastream. However, we find errors in Datastream's total return indices that are mainly caused by price differences and incorrect adjustments for dividends and corporate actions. One of the important weaknesses is that Datastream does not provide time series information about the market segment in which a stock is listed. As a consequence the standard procedure of using portfolio breakpoints from the top market segment cannot be followed. We show that this has important implications when studying the size effect in Germany. To sum it up, we cannot recommend Datastream as the primary data source before 1990. Equity data for the time period after 1990 should be handled with care.

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1 Introduction

Rosenberg/Houglet (1974) argue that “[t]he presence of erroneous data can destroy a research effort and seriously damage the management decisions based upon research.” Random errors in equity data may occur because data is missed or incorrectly recorded. Systematic errors may arise because data such as dividends and stock splits are more carefully collected for large firms than for small firms. Ince/Porter (2006) for example find a higher concentration of data problems in equity data from Datastream for smaller firms. De Moor/Sercu (2013) argue that “[a]mong empirical researchers, it is generally accepted that the probability of data errors is negatively related to firm size.” A possible reason is that large firms are more important from an economic perspective. As a result of omitted dividends and stock splits, returns of small firms are usually systematically underestimated. As a consequence one might not find empirical support for the well documented size anomaly of Banz (1981). Additional problems stem from selection and/or survivorship biases, which also tend to cluster around smaller firms.

Even though, empirical research on the stock market depends on sufficient data quality, an overwhelming part of the economic literature on the cross-section of stock returns does not address data quality directly. This is largely attributed to the focus on the U.S. stock market. Studies for the U.S. market usually rely on the CRSP tape, which is generally perceived as being of high quality.¹ The situation in terms of data quality and availability for non-U.S. markets is quite different. For most countries Thomson Reuters Datastream (Datastream) and Bloomberg represent the only available data sources with broad market coverage and a long historical return time series. Despite the warning of Ince/Porter (2006), equity data from Datastream became quite popular in cross-country studies. For example, McLean et al. (2009), Lee et al. (2009), Campbell et al. (2010), Griffin et al. (2010), Fama/French (2012), and Bartram et al. (2012) employ cross-country equity data from Datastream. Most of these studies are aware of the fact that Datastream data can be prone to errors and, therefore, often screen the data as suggested by Ince/Porter (2006). We generally observe that many recent empirical studies rely on the performance of mechanical screening and winsorizing techniques to address data errors. Very few studies examine the quality of equity data from Datastream and the empirical consequences of using Datastream as the primary data source in more detail. A recent paper by Iqbal et al. (2008) about UK open offers even had to be rewritten because of Datastream’s failure to adjust prices for open offers prior to 2/2002.

¹ The quality of the CRSP tape has been examined by Rosenberg/Houglet (1974), Bennin (1980), Courtenay/Keller (1994), Shumway (1997), Shumway/Warther (1999) and Ince/Porter (2006). In addition, the quality of the CRSP tape is steadily improved by user-initiated corrections. Nevertheless, the CRSP tape is not free of errors. For example, Ince/Porter (2006) argue that “CRSP does not reflect the additional shares or the change in market capitalization until the end of the quarter or fiscal year” for seasoned equity offerings.

Datastream seems to be less popular in countries for which country specific data sets are available. For example empirical studies for the U.S. usually use data from CRSP and COMPUSTAT. UK studies often use the London Share Price Database (LSPD) and an accounting database introduced by Nagel (2001). Studies on the German market as Schrimpf et al. (2007), Artmann et al. (2012a, 2012b) and Brückner et al. (2012) also use country specific data sets. Country specific data sets are, however, rare. Jorion/Goetzmann (1999) find only six markets for which non-Datastream total return time series are available. Dimson/Marsh (2001) obtain country specific return data for 13 countries. They warn, however, that “the quality of the data varies considerably.” For Germany, the Karlsruher Kapitalmarktdatenbank (KKMDB) and a data set maintained by the Centre for Financial Research (CFR) are to our knowledge the only publically available country specific data sets with a sufficient historical time series of German stock returns. The KKMDB is, however, not free of errors and little is known about the quality of the CFR tape. Possibly as a consequence, a growing number of studies on the German stock market use Datastream as the primary data source.

The quality of Datastream has been examined for the U.S. market by Ince/Porter (2006) who document several data errors in Datastream such as coverage issues, return reversals and wrong data for stock splits. Similar problems in Datastream’s data for the UK and their implications are discussed by Andrikopoulos et al. (2007), Espenlaub et al. (2009) and Rossi (2011). Among others Leippold/Lohre (2007) and Schmidt et al. (2011) indicate that Datastream’s data quality varies across countries. Griffin et al. (2010) partially address such issues applying country-specific data filters. Datastream’s data quality for Germany might be comparable to other developed markets such as the UK and France. However, for Germany and most non-U.S. markets we still do not have a clear picture of actual data errors in Datastream and how the use of Datastream data affects the results of empirical studies.

To examine the extent to which equity data from Datastream for the German market is subject to random and systematic errors, we compare two initial data sets for the top segment in Frankfurt with each other. The first data set is our own hand collected data set. The second data set is based on equity data from Datastream. By checking every single data item which is different we create a data set which we believe is accurate, our new data set. This data set we compare with Datastream data to find systematic and random errors in Datastream. In a first step, we check the completeness of the data available from Datastream. We document several severe coverage issues before 1990. For example, Datastream’s coverage of dividends is very low before 1988, increases considerably from 1988 to 1990 and is only sufficient from 1990 onwards. Market coverage, i.e. the number of stocks for which a total return time series is available from Datastream, is incomplete before 1990. To make matters worse, we find a selection bias which results from Datastream’s lower coverage of small firms, and a survivorship bias in Datastream before 1990. As a consequence of these problems, we do not recommend Datastream as the primary data source before 1990. After 1990, coverage problems are

much smaller, i.e. total return and dividend data are available for most stocks listed in the top segment in Frankfurt. A selection bias after 1990 may occur, if German non-voting shares are accidentally removed from the data set. This may happen since Datastream incorrectly classifies German non-voting stocks as preferred stocks. In a second step, we carefully examine the quality of equity data from Datastream for the period from 1990 to 2007. We find random errors in Datastream's time series of the total return indices which are caused by incorrect adjustments for dividends and corporate actions. However, between January 1990 and December 2000 we find many return differentials between the two data sources which are caused by price differences, i.e. Datastream's end of month prices do not match with end of month prices from our data set for Frankfurt. This might have important implications for short term event studies. We also find a considerable number of errors in Datastream's number of shares (NOSH) time series. This error affects estimates of firm's size, value-weight returns and sorts by size. In cross-sectional regressions errors in the NOSH time series may increase the error-in-variable bias. Overall, we find that after 1990 serious data errors are rare for the stocks listed in the top segment of the Frankfurt stock exchange.

However, perfect data accuracy is rarely achieved. The required degree of data accuracy may vary considerably across different application types. To examine the extent to which empirical results for Germany are affected by the use of Datastream data we study the size effect of Banz (1981). We choose the size effect because recent studies by Amel-Zadeh (2011), Hanauer et al. (2012), Artmann et al. (2012a, 2012b) and Brückner et al. (2012) find contradicting results for the size effect in Germany. Our objective is to shed some light on the discussion of the size effect in Germany. This is important, as is it discussed in Germany whether the CAPM should be extended by size or by a factor that represent size. Our empirical test builds on the cross-sectional test procedure of Fama/MacBeth (1973), which has been advocated to study whether the cross-section of returns are related to firm characteristics. We look at the overall period from 7/1975 to 10/2007 and two subperiods, 7/1975 to 6/1990 and 7/1990 to 10/2007. We choose these subperiods for two main reasons. First, before 1990, Datastream is subject to systematic errors and any empirical results, thus, difficult to interpret. After 1990, however, the quality of equity data from Datastream is sufficient, especially with respect to market coverage. Second, looking at these subperiods also allows us to replicate the results of Brückner et al. (2012), who document a strong reverse size effect in Germany during 7/1990 and 10/2007.

In line with Brückner et al. (2012) we find that the size effect is not stable over time in Germany. This may explain the different results of previous studies for the size effect in Germany. The variation in prior results may also be caused by the use of different data sets, i.e. previous studies cover different stock exchanges and/or market segments. To understand how the inclusion of German stocks from different German stock exchanges and market segments may affect empirical results, we look at different sets of size sorted portfolios when we study the size anomaly in Germany. Using equity data

from Datastream we find no size anomaly in Germany during the period from 1990 to 2007 when we follow common practice to create test portfolios, i.e. we select data for all German equities and form size decile portfolios, whereby we assign each decile the same number of securities. These results are in support of those reported by Artmann et al. (2012a) who also find no evidence of a size effect for the period from April 1988 to December 2006. However, we additionally form size deciles based on all German equities from Datastream using size breakpoints for the top segment in Frankfurt. Similar sorting schemes have been advocated by Fama/French (1992, 2012) to avoid sorts that are dominated by the plentiful small and tiny firms. Applying this sorting scheme we obtain results similar to those by Brückner et al. (2012) and Hanauer et al. (2012), i.e. a reverse size effect during 7/1990 to 10/2007 emerges. The reverse size effect is strongest when we restrict the sample to the stocks listed in the Amtlicher Markt in Frankfurt. Overall, our data set and data from Datastream yield the same results for the top segment of the Frankfurt stock exchange. This suggests that results for the size effect are quite robust to data errors. Finally, we sort all German equities from Datastream into two portfolios, small and large, using median size as breakpoint. Looking at the stocks in each portfolio separately, we find a regular size effect among small stocks and a reverse size effect among large stocks for the period from 7/1990 to 10/2007.

Our paper distinguishes from related papers such as Ince/Porter (2006) and Schmidt et al. (2011) in several ways: First, we carefully examine errors in Datastream using a unique data set for the top segment of the Frankfurt stock exchange. This is in light of Ince/Porter (2006), Andrikopoulos et al. (2007), Espenlaub et al. (2009) and Rossi (2011) who examine the quality of Datastream's U.S. and UK tape. However, we are not aware of any paper that examines Datastream's data quality for Germany or any markets other than the U.S. and the UK. Second, we summarize important German peculiarities that must be considered more carefully in empirical studies that include the German market. Third, we check whether errors occur more frequently among small firms compared to large firms. Our results for the top segment in Frankfurt suggest that with respect to market coverage and coverage of dividends this is only the case before 1990. Finally, we show how empirical results for the size effect in Germany vary with the data set and the grouping procedure.

The remainder of the paper is organized as follows. In Section 2 we briefly discuss German peculiarities that need to be considered when creating a data set for the German market. In this section we also describe available data sets for the German market in more detail. In Section 3, we start evaluating the quality of equity data from Datastream and provide arguments for not using it before 1990. Since we currently do not recommend Datastream as the primary data source before 1990, we focus primarily on the period from 1/1990 to 10/2007 in Section 4 to investigate whether Datastream's data quality is sufficient for standard academic use after 1990. In Section 5, we compare market-wide average rates of return across different data sets by replicating popular German all-share stock market indices like the CDAX and the DAFOX. The main focus of Section 5 is to show how empirical results

for the size effect in Germany vary across different data sets for the German stock market. Section 6 concludes.

2 Equity Data for the German Market

2.1 German Peculiarities

Throughout the period from 1973 to 2007 eight stock exchanges existed in Germany (currently seven). During most of this period Frankfurt was by far the most important stock exchange in Germany. Historically, each stock exchange has three market segments (since November 2007 only two), the top segment was the Amtlicher Markt (now the Regulierter Markt). Most stocks are listed at more than one stock exchange and in different market segments. A firm's stock is not necessarily traded on all stock exchanges every day. Often only indicative prices are available. Usually stock prices differ across stock exchanges, and price differences are typically larger for small and illiquid stocks. As a consequence monthly returns vary considerably across stock exchanges. Long term holding period returns are, however, less affected by this issue. Nevertheless, possible consequences are average returns (arithmetic mean of monthly returns) that are down-/upward biased and measurement errors in risk measures like beta and standard deviation. To alleviate such problems Brückner et al. (2012) choose prices based on turnover, i.e., they select the price from the stock exchange with the highest turnover on the last trading day of the month. In most cases this is Frankfurt, but there are cases where turnover is higher on other stock exchanges, this is usually the home stock exchange (Heimatbörse).

Empirical studies on the cross-section of German stock returns differ with respect to the stock exchanges and the market segments they include. Stehle (1997), Schulz/Stehle (2002), Schrimpf et al. (2007), Ziegler et al. (2007) and Brückner et al. (2012) focus primarily on the top market segment of the Frankfurt Stock Exchange, the Amtlicher Markt in Frankfurt. Amel-Zadeh (2011) and Hanauer et al. (2012) consider all stocks listed in the German composite index, thus they also focus on Frankfurt, but include firms from the lower market segments, the Geregelter Markt in Frankfurt (*5/1987-†10/2007) and the Neuer Markt (*3/1997-†6/2003). Stocks from the “unofficial” market segments (Open Market, formerly known as Freiverkehr, Geregelter Freiverkehr and Ungeregelter Freiverkehr), which are generally perceived as over-the-counter markets, are usually not considered. Studies that do not explicitly distinguish between the different market segments are Oertmann (1994), Schlag/Wohlschließ (1997), Wallmeier (2000) and Elsas et al. (2003). Some studies for the German market do not clearly distinguish between the different market segments and are as a result probably subject to an ex post selection bias, e.g. Artmann et al. (2012a, 2012b).

The stocks listed in the lower market segments are usually tiny compared to the stocks listed in the top segment. Including the stocks from the lower market segments might increase regression fits due to the higher number of firms per portfolio. However, as a consequence empirical results are dominated by the plentiful, but economically less important tiny stocks from the lower market segments. To

overcome this problem one could form portfolios based on breakpoints from the top market segment. This is comparable to commonly applied sorting procedures for the U.S. market. Among others Fama/French (1992) form portfolios based on NYSE breakpoints to avoid portfolios that are dominated by Amex and NASDAQ firms. However, this sorting scheme is difficult to replicate without reliable information about the constituents of German market segments. Alternatively, to avoid empirical results that are dominated by small firms one may use size breakpoints that are percentages of aggregate market capitalization as in Fama/French (2012). Nevertheless, because of different legal supervision, admission, and listing requirements between the German market segments, empirical results might be subject to a market microstructure effect as described by Reinganum (1990) and Loughran (1993) for the U.S. In addition empirical results might be affected by an IPO effect. The lower market segments, especially the Neuer Markt, attracted a considerable number of IPOs. Out of ca. 623 IPOs (in our IPO database) that occurred at the stock exchange in Frankfurt during 1/1988 and 10/2007, only 152 occurred in the Amtlicher Markt. Most IPOs, 294, occurred in the Neuer Markt during 3/1997 and 3/2003. For the Geregelter Markt we observe 111 IPOs and for the unofficial market segment 71.

In Germany, we generally distinguish between Stammaktien (common stocks) and Vorzugsaktien (non-voting stock). Both stock classes are considered as equity and should be included in a sample of German equities. However, Vorzugsaktien are usually translated with preferred stocks. Datastream typically adds 'PREF.' or 'PF.' to the name of German non-voting shares classifying them as preference shares. Ince/Porter (2006) suggest removing preferred shares. This recommendation is usually followed by international studies that include the German market, e.g., McLean et al. (2009), Lee et al. (2009) and Hou et al. (2011). Thus, German non-voting shares are often incorrectly removed from the data set because of Datastream's incorrect classifications. Some German firms have only their non-voting shares listed in Frankfurt, e.g., Porsche AG. Non-voting shares are frequently listed prior to the common shares, e.g., Hugo Boss AG and ESCADA AG. For some firms the common shares are listed first, e.g., Deutsche Lufthansa AG and HeidelbergCement AG. Usually the two share classes are listed at the same time, e.g. Volkswagen AG. Since December 18th, 2009 the non-voting shares of Volkswagen AG are also represented by the most prominent German blue chip index, the DAX. Approximately 80 firms (ca. 16% of all firms) that are listed in the Amtlicher Markt in Frankfurt during 1970 and 2007 have both share classes listed at the same time. Of these, at least 30 firms converted their non-voting shares into common shares. Removing non-voting shares from the data set, thus, introduces coverage issues. Sorts on firm size, book-to-market ratios and value-weight returns, where implicitly less weight is assigned to dual class firms, will be affected unless all stock classes of a firm are considered to estimate firms' market capitalization.

Some stock exchanges, for example the NASDAQ, delist stocks, which have a permanent stock price of less than \$1.00. The Deutsche Börse AG also tried, but failed to delist penny stocks from the Neuer

Markt. Ince/Porter (2006) “drop all observations in both the TDS and CRSP samples when the end-of-previous-month price is less than \$1.00.” Schmidt et al. (2011) “delete all so-called ‘Penny-stocks’ with prices less than one unit of the domestic currency.” These screens also remove large German firms such as Infineon Technologies AG during 12/2008 and 3/2009 when the firm’s share price was less than €1, throughout this period the market capitalization was typically well above €300 mln. Another example is YMOS AG, the price of the firm’s common shares is typically below €1.00 during the period from 1/2004 and 10/2007. At the same time the market value is on average €30 mln. which is larger than of many smaller stocks with prices well above €1.00. We classify stocks with prices below €1.00 and market value of equity below €5 mln. as penny stocks. However, we find no penny stocks in the top segment of the stock exchange in Frankfurt before 10/2000. From 10/2000 to 10/2007 we classify on average 5% of the stocks listed in this market segment as penny stocks. The fraction of penny stocks increases to 10% if we add the second market segments, the Geregelter Markt in Frankfurt, and the Neuer Markt to the sample.

2.2 German Equity Data from Datastream

In addition to the above discussed issues, five more problems have to be considered when Datastream is used as the primary data source for German stocks. First, Ince/Porter (2006) find “that the full time series of classification variables often reflect only the most current value.” For Germany this means that stocks that were initially traded on a regional stock exchanges and later switch to Frankfurt are classified as “Frankfurt” securities or vice versa. In studies that include all German equities this is not a problem, unless coverage and data quality are lower for the regional stock exchanges. Studies that only look at the Frankfurt stock exchange might, however, be subject to a survivor ship bias since only successful firms eventually start their listing in Frankfurt.

Second, until recently Datastream did not distinguish between the German market segments. Currently, Datastream provides static information on the current constituents of the market segments. Time series data of the market segments a stock was listed in is to our knowledge not available yet. This information is, however, important since OTC stocks are typically not included in studies on the U.S. market and in most studies on the German market. To avoid a survivorship bias in studies that only look at the top market segment (or specific market segments), it is important to remove firms from the sample for the period they were listed in the lower and unofficial market segments (OTC markets). This is, however, difficult to achieve as Datastream does currently not provide a time series for the market segment a stock was listed in. As a consequence of this weakness one cannot “easily” distinguish between stocks from the top segment and the lower segments throughout time. This may lead to biases, since coverage and data precision may be different for the different segments of the German stock market. Following our recommendation Datastream recently introduced new data types, SEGM and SEGN, which provide static information about the market segment a stock is currently listed in. However, this information does not allow distinguishing between stocks from the Amtlicher

Markt, Geregelter Markt and Neuer Markt in Germany, because most of these stocks are listed in the Regulierter Markt since 11/2007.

Third, Datastream classifies Frankfurt's floor trading as the primary market in Germany, XETRA is usually classified as the secondary market. XETRA was introduced in 11/1997 (replacing the IBIS trading system), since then most trading in stocks takes place in XETRA. Before 11/1997 XETRA prices are not available. We find, however, that Datastream's XETRA time series report prices and total returns before the XETRA trading system was introduced. The classical floor trading ended on May 23rd, 2011. Datastream usually provides several DSCDs for most stocks. These DSCDs vary across stock exchanges and currencies. Each DSCD may, however, be considered as a unique identifier for a combination of a stock, stock exchange and currency.² We select data primarily from the time series for Frankfurt, data from XETRA or other German stock exchanges are considered only if data (RI, NOSH, UP) is not available for Frankfurt. The research lists, except for the defunct lists, usually include Datastream identifiers (DSCD) for Frankfurt only. DSCDs for XETRA may only be obtained by searching Datastream for all German equities manually. Searching Datastream for all German equities, however, requires screens that ensure that only one DSCD for each stock is included in the final sample.

Fourth, the available research lists (FGER1, FGER2, FGERDOM, FGKURS) and defunct lists (DEADBD1 to DEADBD6) for the German market are incomplete. This issue is also pointed out by Ince/Porter (2006) for the U.S. market. We find 246 additional stocks searching Datastream for all German equities (filter: status=all, market=Germany, instrument type=equity, see Table 1). This extended search requires no additional information, but improves coverage considerably.

Fifth, Datastream reports generally one DSCD per stock class and stock exchange. For some stocks Datastream reports multiple identifiers for the same stock exchange, for example to distinguish between different currencies. Such issues are not necessarily recognized by common screening techniques unless local security numbers are additionally merged. However, local security numbers do change over time or might not be available from Datastream for all stocks.

We create two data sets using equity data from Datastream. The first data set is created following common practice in international studies as described in detail by Ince/Porter (2006). We select securities from Datastream's country specific active and defunct research lists. Additionally we search

² Datastream recently introduced time series for specific stock exchanges. Stock exchange specific time series of un-, adjusted prices and total returns can be retrieved from Datastream using the data type plus the exchange code. For Frankfurt this would be for example UP.FF, P.FF and RI.FF. These time series for Frankfurt, however, also use data from other stock exchanges when data from Frankfurt is not available. So far we have little experience with these new data types. However, checking the data for Bausch AG reveals that Datastream's UP.FF time series contains prices for the period before April 10th, 1996, a period the stock was not listed in Frankfurt. The time series of unadjusted prices for Munich, UP.MU, reports the last price change on August 5th, 1996, long before the stock was delisted in Munich.

for all German equities. Next, we screen these lists for non-equity securities such as warrants, exchange traded funds, etc. However, we do not remove non-voting stocks which are incorrectly classified as preferred stocks by Datastream. Consolidating the stocks from the aforementioned lists yields a sample of 1,747 German stocks that are listed in Germany during 1/1973 and 10/2007 (see Table 1). The second data set includes stocks only for the exact time period during which they are listed in the top segment in Frankfurt from 1973 to 2007. This data set is created using our proprietary constituent list for this market segment. The quality of this data set for the top segment in Frankfurt is examined in Section 4.

[Table 1]

Table 2 provides an overview of data types from Datastream that we consider in our study. For each stock we download the daily time series of unadjusted prices (UP), adjusted prices (P), and total return index (RI). We also consider the RZ time series, which is an alternative to the RI time series of total returns. In some cases the RZ time series tends to perform slightly better with respect to adjustments for dividends than the RI time series. However, we use primarily the RI time series as most of the empirical papers do. We retrieve two decimal places for the RI time series (Datastream's default setting).³ We also obtain time series data on dividends (UDDE) and capital adjustments (AX). We convert prices and dividends reported in DM into EURO.⁴ We download the number of shares (NOSH) to estimate companies' market value of equity (UP multiplied by NOSH). Looking at NOSH allows us to examine the accuracy of adjustments of the number of shares following corporate actions. Monthly rates of return are calculated from the total return indices using end of month levels.

[Table 2]

2.3 Our Data Set for the Top Segment in Frankfurt

The data collection for our data set was initiated by Richard Stehle in 1978. The first version of the data set was completed in 1989 and described by Stehle/Hartmond (1991). The original data set was expanded among others by Schulz/Stehle (2002) and Brückner et al. (2012) until October 2007.⁵ The data was obtained from the fact books: Hoppenstedt Kurstabellen (until 1998), the Hoppenstedt Aktienführer (1998-2010), Saling Aktienführer (until 1995), and Handbuch der Deutschen

³ Ince/Porter (2006) argue that rates of return based on RI with two decimal points are erroneous, especially when RI values are low. This is correct. However, Datastream allows to retrieve more than two decimal points by using the DPL# function, e.g. using DPL#(X(RI),4) instead of RI yields four decimal places. We have to admit that we were not aware of this function. However, for the stocks listed in the top segment of the Frankfurt stock exchange we find only 635 observations where rates of return based on the RI time series differ by more than 0.5% from returns based on the DPL#(X(RI),9) time series. Of these 493 are for penny stocks. Thus our results are not driven by the number of decimal places in RI.

⁴ For some stocks the UDDE time series first reports dividends in DM and later in EURO. We use the DCRE time series, which provides the currencies for dividends reported by the UDDE time series, to convert DM dividends into EURO.

⁵ Previous versions of the data set were applied among others by Stehle (1997), Dimson/Marsh (2001), Schrimpf et al. (2007) and Ziegler et al. (2007). The current tape is also applied by Brückner et al. (2012) and Brückner/Stehle (2012).

Aktiengesellschaft (1953-1994). Additional electronically available data sources are the Karlsruher capital market database (KKMDB, 1970-2007), the *Börsenzeitung* (10/1998-2007), XETRA Newsboard (1998-2007), DGAP (1998-2007), and if necessary Datastream (1973-2007).⁶ Our data set contains all necessary data to calculate monthly total rates of return, including the last price of each month, dividends, pure stock splits, stock dividends, rights issues, reverse stock splits and other financial benefits to shareholders. Finally, the data set contains the number of shares outstanding which is required to estimate stocks' market capitalization of equity and value-weight portfolio rates of return.

The initial data set includes all German firms of which at least one class of shares was listed in the top segment of the Frankfurt Stock Exchange the "Amtlicher Markt" in Frankfurt between December 1953 and October 2007.⁷ The data set is restricted to the Amtlicher Markt in Frankfurt for two main reasons. First, high quality data for the other stock exchanges are currently not available to us. Second, many empirical studies for the German market also focus solely on the top segment in Frankfurt. To avoid any selection bias or survivorship bias, firms are only included for the period for which they were actually listed in the Amtlicher Markt in Frankfurt. IPOs and other new listings are added to the data set at the end of the month of their first listing in this segment. We assume that the data set includes all German stocks for the entire time they are listed in the Amtlicher Markt in Frankfurt. Furthermore, the data set is restricted to common and non-voting stocks only. Profit participation bonds (*Genussscheine*), "Restquoten", and firms that are liquidated, but still exchange listed are not considered.

Stock prices are mainly taken from the Stehle/Hartmond (1991) database (until 12/1995), the KKMDB (available from 1/1974), and the *Börsenzeitung* (available from 11/1999). Occasionally, we fill up missing prices with stock prices from the Hoppenstedt Kurstabellen (available until 12/1998), Datastream (available from 1973), and Ariva (available from 1/1990). We primarily select prices from Frankfurt (floor trading). Only if we cannot find an end of month trading price for Frankfurt we use prices from other stock exchanges. The market value of equity is generally estimated as the product of the stock price and the number of shares outstanding as of the end of each month. The number of shares outstanding was initially obtained by Stehle/Hartmond (1991) and expanded by Schulz/Stehle (2002) and Brückner et al. (2012) using the aforementioned data sources. In addition, we cross-checked the number of shares using information on stock splits, stock dividends, rights issues and reverse stock splits.

⁶ The data from the *Börsenzeitung* is available from <http://wpi.boersen-zeitung.de>. The XETRA Newsboard is provided by the Deutsche Börse AG under <http://deutsche-boerse.com>. Data from DGAP is available from www.dgap.de.

⁷ The data set extends back to 1938 for the largest German stocks which would have been included in the DAX.

We calculate monthly rates of return from the perspective of domestic minority shareholder following the procedure described by Stehle/Hartmond (1991). Thus, we adjust monthly rates of return for share reallocations from majority to minority shareholders, and dividends which are only distributed to minority or free shareholders. To make our returns more comparable to those from Datastream we do not take tax refunds (Körperschaftsteuergutschrift) into account. However, tax refunds represent a major return component to German shareholders from 1977 and 2000 and should, therefore, be included as in Brückner et al. (2012). We estimate adjustment factors for stock dividends, pure stock splits, and reverse stock splits according to Sauer (1991). We also take the chronological order of events (in case of multiple events on a single day or during the month) into account when we calculate monthly rates of return.

The majority of seasoned equity offerings in Germany are rights issues. This complicates return calculations in the following way. First, before 1995 we use the first trading price of the subscription rights (if available) as Stehle/Hartmond (1991) and Göppl (1995). We use the theoretical value of subscription rights for the period from 1996 to 2007, because for this period the actual prices are not (yet) available to us. Datastream's total return time series and most stock indices as for example the CDAX of the Deutsche Börse AG are also adjusted for subscription rights using their theoretical values. According to Lorenz/Röder (1999) and Dorfleitner/Röder (2002) the actual prices of the subscription rights are systematically lower than their theoretical values, for the Amtlicher Markt on average -11.01 to -11.77%. As a consequence rates of return are systematically upward biased when we use the theoretical value of the subscription rights. This has to be carefully considered in event studies that analyze seasoned equity offerings. Second, we assume that dividends and the proceeds from selling the subscription rights are reinvested at the end of the month. Datastream and most index provider assume that dividends and proceeds are reinvested immediately at the (first or closing) price ex dividend or ex subscription rights. However, the resulting return differentials are random and should have little implication for long-term rates of return.

2.4 Other Data Sets for the German Market

The University of Karlsruhe provides equity data for the German market, which has become known as the Karlsruher Kapitalmarktdatenbank (KKMDB). The KKMDB is documented by Bühler et al. (1993) and Herrmann (1996). The time series of prices extend back to 1960 for 100 German stocks. Data for all German stocks is provided starting in 1974. The major strength of the KKMDB is that it provides stock prices for all German stock exchanges and market segments. The KKMDB is frequently applied as a starting point to create an initial data set. For example, Artmann et al. (2012b) obtain daily stock prices from the KKMDB which they adjust for dividends, stock splits, and equity offerings using data from the KKMDB and the Saling/Hoppenstedt Aktienführer. In other words, they implicitly check and correct the data from the KKMDB for missing return components.

The data set available from the Centre for Financial Research (CFR) at the University of Cologne is described in detail by Artmann et al. (2012b). The CFR data set consists of various equal-weight one-dimensional and two-dimensional sorted portfolios. Portfolios are formed sorting stocks by size, book-to-market, beta and momentum. Artmann et al. (2012b) “[...] include all firms listed on the market segments ‘Amtlicher Handel’ or ‘Neuer Markt’. In addition, [they] consider stocks of firms listed on ‘Geregelter Markt’ if they were listed on ‘Amtlicher Handel’ or ‘Neuer Markt’ at any time during [their] sample period.” Brückner/Stehle (2012) indicate, firms that made it from the Geregelter Markt to the Amtlicher Markt were among the most successful firms (“winners”). As a consequence, the data set is probably subject to an ex post selection bias. To our knowledge Artmann et al. (2012a, 2012b) are so far the only empirical studies that use this data set. We will consider the data from the CFR in our empirical section on the size effect when we compare average portfolios returns.

Another data set for the German market is provided by Kenneth R. French. French’s data library contains data for value-weight portfolios of German stocks. Portfolios are formed by book-to-market, earnings-price, cash earnings to price and dividend yield. However, the data set covers only a fraction of the German stock market before 2008 and might thus be subject to a selection bias. Finally, equity data for the German market is also available from Bloomberg.

3 Quality of Datastream Data before 1990

Currently, equity data (especially total return data) from Datastream is available for Germany from 1973 onwards. From 1973 to 1990 338 stocks were listed in the top segment in Frankfurt according to our data set. Searching Datastream for these stocks yields a list of 300 Datastream identifiers (DSCD). Table 3 shows that Datastream misses 38 identifiers for the period from 1973 to 1990. Thus Datastream misses identifiers for 11.34% of the stocks.

[Table 3]

For each stock we download the time series of the total return index, RI, for each security and compare its return history available from Datastream with the return history from our data set. During the periods from 1973 to 1990 a total return time series is available only for 85.50% of the stocks listed in the top segment of the Frankfurt Stock Exchange. A complete total return time series is available only for 61.42% of the stocks listed in the top segment. Datastream’s RI time series is incomplete for 82 stocks. On average the RI time series misses total return data for the first 132 listing months during 1973 and 1990. After 1990 a complete total return time series is available for 97.92% of the stocks.

We also find a survivor ship bias in Datastream before 1990. Only 17 DSCDs are available for the 54 stocks that ceased listing in the top segment in Frankfurt during 1973 and 1990. Total return data is available for 11 stocks only, for two stocks the available total return time series is incomplete. In the words of Ince/Porter (2006) “having data for a country in a particular year does not imply that coverage in that year is complete”. Finally, Table 8 (which is discussed in more detail in Section 5)

indicates that between 1973 and 1990 Datastream's coverage of small firms is considerably lower than its coverage of large firms, thus introducing a selection bias. Table 4 shows that we are only able to merge 67.89% of the monthly returns for the period from 1/1973 to 12/1989. However, 3,580 (11.62% of the merged returns) monthly returns do not match, i.e. monthly return differentials are larger than 1% across the two data sets. The number of "large" return differentials is considerably lower, only 0.96% of the return differentials are larger than 5%.

[Table 4]

We also match dividend data from Datastream and our data set for the firms listed in the Amtlicher Markt between January 1973 and October 2007. Datastream's data type definition of the total return index states that "detailed dividend payment data is only available on Datastream from 1988 onwards." Table 5 shows that between 1981 and 1987, the UDDE time series on average provides dividend data for less than 12.0% of the dividend paying stocks. During 1988 and 1990 dividend coverage of the UDDE time series increases considerably, from 53.5% to 97.0%. As a consequence, we consider dividend coverage of the UDDE time series insufficient for most stocks before 1990. Since 1990 dividend coverage is well above 98% (see Table 5). Even though dividend coverage of the UDDE time series is insufficient before 1990, the total return indices, RI, are still adjusted for dividends in many cases when there is no entry in the UDDE time series. For most stocks Datastream adds an increment of $1/260^{\text{th}}$ of the annualized dividend yield to the RI time series per day until 1988. As a consequence, we find many return differentials over the month of a dividend payment between Datastream and our data set before 1990. These return differentials are fairly close to average dividend yields from our data set and thus well explained by incorrect dividend adjustments. We are able to match 1,555 dividend yields on an annual basis (monthly dividend yield times 12) across the two data sets for the period from 1973 to 1990. During this period 3,283 dividend payments occurred, of these we are only able to match 351 dividend yields on a monthly basis.

[Table 5]

For the period from 1/1973 to 12/1990 Datastream's AX time series provides only 515 (71.2%) instead of the required 723 adjustment factors. Coverage of corporate actions also increases over time. The low coverage of corporate actions before 1990 is mainly caused by Datastream's insufficient market coverage (see Table 3). For most missed corporate actions in the AX time series the RI time series is also empty.

To summarize the problems, we document important issues of coverage in Datastream before 1990. During the period from 1973 to 1990 we find market coverage with respect to the stocks listed in the top segment of the Frankfurt stock exchange to be insufficient. Dividends are generally not included in the UDDE time series. The total return time series are not properly adjusted for dividends. For these reasons we cannot recommend Datastream as a single source of equity data for the German market before 1990. Market coverage, in terms of available Datastream identifiers, DSCDs, dividend

coverage, and availability of total return data increases considerable over time. The coverage issues before 1990 have implications for various types of empirical studies. For example you cannot rely solely on equity data from Datastream to measure the performance of IPOs in Germany before 1990. Since total returns are inappropriately adjusted for dividends, abnormal returns might also be biased, especially on the ex-dividend day. Long-term market-wide rates of return are, however, not affected as indicated in our empirical section. For the aforementioned reasons we do not examine the quality of Datastream's equity data in detail before 1990.

4 Quality of Datastream Data after 1990

Datastream's equity data for the period from 1990 onwards is of high quality with respect to market coverage and coverage of dividends. For example, we find 575 Datastream identifiers for the 576 stocks in our data set for the period from 1990 to 2007 (see Table 3). However, random errors do occur. To provide a clear picture of actual errors in Datastream's time series we carefully compare equity data from Datastream with data from our data set for the top segment in Frankfurt. We focus on the period from 1/1990 to 10/2007, when the Amtlicher Markt in Frankfurt was closed. Comparing returns from Datastream's time series of the total return indices, RI, with returns from our data sets reveals many return differentials that are larger than 1%. Most return differentials are caused by price differences between the two data sets. We also examine whether Datastream's total return indices, RI, is adjusted for dividends and corporate actions such as pure stocks splits and subscription rights. We also document a significant number of errors in Datastream's NOSH time series. Some of these errors are of very high magnitude. However, we do not find systematic errors in the RI and NOSH time series with respect to capital actions. We use various data sources to evaluate mismatches between Datastream and our data set. Overall, we find many errors in Datastream's equity data, examples for specific errors are available upon request.

4.1 Prices and Rates of Return

Comparing monthly rates of return for the period from 1990 to 2007 we find 2,406 (3.29% of merged returns) return differentials that are larger than 5%, 297 (0.41%) are even larger than 15%. Most return differentials are caused by differences in the end of month closing prices; less return differentials in percent occur when we look at non-overlapping quarterly or annual rates of return (geometric means of monthly returns).⁸ Table 4 shows that the relative number of return differentials is highest for the period from 1/1990 to 12/2000, when only 67.54% of the returns from the two data sets match (return differentials are less than 1%). This problem is persistent across all size classes during this period. The relative number of return differentials is considerably lower for the preceding period from 1/1973 to

⁸ Ince/Porter (2006) observe for the U.S. that closing prices deviate between CRSP and Datastream.

12/1989 where 88.38% of the available returns match. For the succeeding period from 1/2001 to 10/2007 we are able to match 95.53% of the returns.

A large number of the return differentials are related to three issues. First, in many cases the last change in Datastream's UP and RI time series occurs before the last trading day in a month. As a consequence, 839 monthly return differentials larger than 5% emerge for the period from 1990 to 2007. Most return differentials (785 out of 839) occur for the period from 1/1992 to 12/2000. This coincidence with the time period when Datastream's German equity data was provided by Telekurs.⁹

Second, looking at 4,118 price differences for the period from 11/1999 to 10/2007 we find that both the KKMDB and Datastream frequently report prices that do not match closing prices for Frankfurt as reported by the *Börsenzeitung*. For 2,461 observations we find that Datastream's price matches with the price from the *Börsenzeitung*. For 1,110 observations the price from the KKMDB matches with the *Börsenzeitung*. For 691 observations we cannot match prices from Datastream nor the KKMDB with prices from the *Börsenzeitung*, mainly because prices are not available from the *Börsenzeitung*. Out of these 691 observations, however, we can compare 323 with prices from Ariva; 322 prices match with the KKMDB and only one matches with Datastream.

Third, the RI time series for Frankfurt is often based on stock prices from stock exchanges other than Frankfurt. This frequently (not always though) coincides with situations where only indicative prices are available for Frankfurt. Originally, we assumed, DSCDs for Frankfurt will yield time series data that are based on stock prices from Frankfurt only.¹⁰ We also find time series of unadjusted prices where Datastream's end of month prices hardly match any prices from the KKMDB. Some prices match those reported by Lang & Schwarz AG (OTC market). For many stock prices, however, it remains unclear to us where Datastream's prices come from.

We also find return differentials that seem to be related to technical problems. For example, the RI time series for Neckarwerke Stuttgart AG varies considerably depending on the start date used to download the data. Using January 1st, 1965 (or before) as the start date introduces a severe error to the time series for the period from March 22nd to May 19th, 2003. During this period the RI time series rapidly increase from 835.66 to 21,849,060, whereas the correct RI time series (based on a start date after January 1st, 1965) increases only to 1076.92. As a consequence of this programming error, return estimates are heavily upward biased. Common screening techniques might not recognize such issues.

⁹ Datastream's data stems from the following data vendors: (1) pre 1990 from BDZ, (2) 1990 - 2002 from Telekurs, (3) 2002 - 2009 from Global Topic, and (4) 2009 onwards from Datascope.

¹⁰ Datastream's stock exchange specific time series of un-, adjusted prices and total returns also use data from other stock exchanges when data from Frankfurt is not available (see FN 3 for details).

We originally expected to observe a higher concentration of return differentials among small, illiquid stocks (especially for those with very low free float) and penny stocks. The literature often suggests that data quality is lower for small stocks. The prices of small stocks also usually vary considerably across stock exchanges. However, the number of absolute return differentials larger than 1% does not change considerably across size deciles for the period from 1990 to 2007. In unreported results we observe ca. 4,543 return differentials for the smallest stocks (size deciles D01 to D03) and ca. 4,183 for the largest stocks (size deciles D08 to D10). However, when we look at return differentials that are larger than 5%, we find more return differentials among small stocks (1,045) than for large stocks (296). Most return differentials are, however, solely caused by price mismatches between the two data sets.

We also examine the performance of one of the screening techniques suggested by Ince/Porter (2006). They suggest removing rates of return that are higher than 300% and reversed within one month. Applying this technique to a list of all German exchange listed equities during the period from 1/1990 to 10/2007 yields only a single instance, a penny stock with rates of return of -96.25% in 5/2003 and 313.41% in 6/2003. Altogether, we observe only 48 rates of return that are higher than 300%, out of these 20 are for penny stocks. For example, the stock price of ISION Internet AG actually increased during 4/2003 from €0.21 to €4.00. We also screen for rates of return higher than 100% that are reversed within one month. Screening for such return yields a list of 161 observations. Again most return reversals (ca. 91) are observed for penny stocks, which are usually not considered in empirical studies. Some of the remaining 70 instances are indeed caused by errors in the RI time series.

4.2 Dividend Coverage

During the period from 1/1990 to 10/2007 we adjust monthly returns for 4,741 dividend payments. We are unable to merge 52 (1.10%) dividends with Datastream's UDDE time series for the following reasons:

- Datastream's UDDE time series misses pure bonus payments in three cases.
- One dividend is missed because we could not find the stock in Datastream.
- Forty-eight dividends are not included in the UDDE time series for the Frankfurt Stock Exchange.

In addition, we find 137 (2.89%) dividend payments (incl. bonuses) that differ in their value across the two data sets by more than €0.01. For these 137 observations UDDE understates the actual dividend payment considerably, on average by €1.67. This difference is mainly caused by missing bonuses in Datastream's UDDE time series. Bonus payments are often substantial in Germany; frequently they are higher than the regular dividend. Altogether, the UDDE time series misses ca. 33 bonus payments completely and only partly reflects 21 bonuses. Most bonus payments are, however, fully reflected by the UDDE time series. We also find errors in the UDDE time series that seem to be caused by

adjustments for corporate actions. In some cases we find that the DCRE time series reports the wrong currency for a dividend in the UDDE time series.

Errors in the UDDE time series do not necessarily imply errors in the total return time series. Hence, we also check whether total returns, RI, are adjusted for dividends by comparing dividend yields (*Div.-Yield*) across the two data sets. We estimate dividend yields for our data set as

$$Div.-Yield = \frac{Div_t}{UP_{t-1}} \quad (1)$$

where Div_t is the dividend (incl. bonuses) from our data set on date t and UP_{t-1} is Datastream's last cum dividend price on trading day $t-1$. We use UP to ensure that differences in dividend yields are not caused by price differences across the two data sets. We compare the first measure with

$$Div.-Yield^* = \frac{RI_t}{RI_{t-1}} - \frac{P_t}{P_{t-1}} \quad (2)$$

where the dividend yield is derived as the difference in the change in the total return indices, RI, and the change in adjusted prices, P. We additionally estimate both measures on a monthly basis using end of month prices. To verify whether dividends are fully reflected by the total return time series we simply need to compare the two dividend yield measures across the two data sets.

Out of 4,741 observations 608 (12.82%) dividend yields do not match during 1990 and 2007, i.e. the difference in dividend yields across the two data sets is on average larger than half a percentage point. Thus, there are problems with dividend yields, even though the UDDE time series reports the correct dividend payment. The 608 problems are caused by:

- We could not check dividend yields for 7 dividend payments because total return indices or adjusted prices are not available.
- 32 dividend yields are too high.
- 82 dividend yields are too low. Of these 21 dividend yields based on Equation (2) are negative, which is implausible. Dividend yields range between -60% and -100% for 20 observations. These negative yields are mainly caused by errors in the time series of adjusted prices, not in the total return index time series.
- 494 dividends seem not to be considered by the total return indices, i.e. the dividend yield from Equation (2) using daily data are virtually zero.

We check whether the above reported 608 mismatches in dividend yields are caused by inappropriate adjustments. Looking at monthly dividend yields (based on end of month prices) result in 235 dividend yields that are fairly close to the “true” dividend yield. Another, 70 monthly dividend yields need to be multiplied by 12 in order to be close to the true dividend yield. This indicates that in some cases Datastream continues to add an increment of $1/260^{\text{th}}$ of the annualised dividend yield to the RI time series, even though detailed and correct dividend data is provided by the UDDE time series. To

mitigate such problems one can alternatively use the RZ time series, which for Germany works slightly better in some cases than the more commonly used RI time series.

Looking at monthly returns for the 608 above identified problems, we find that 183 returns (based on RI) match with our return time series, i.e. the difference in returns is less than half a percentage point. Incorrect dividend adjustments, missed dividends and price differences across the two data sets account for 425 return differentials that are larger than 0.50%. Based on our results, we do not recommend techniques as proposed by Schmidt et al. (2011), who suggest dividing certain dividends by a factor of ten to recalculate returns for the German market. Finally, we find the quality of Datastream's dividend data to be homogeneous across size sorted portfolios (results not shown). Before 1990 the UDDE time series misses many dividend payments across all size classes. After 1990, the quality of the UDDE time series is generally high, but not free of errors. Thus, with respect to dividends we cannot support conventional wisdom that data quality is higher for larger than for smaller stocks.

4.3 Coverage of Corporate Actions

To identify errors in the RI time series that are related to corporate actions, we again compare stocks' monthly rates of return. Here we focus on month during which corporate actions occurred. We additionally look at the AX time series, which provides adjustment factors for corporate actions. Altogether, Datastream's AX time series provides 899 (90.8%) out of 1,034 adjustment factors for the period from 1/1990 to 10/2007. Table 6 summarizes information about the 1,034 corporate actions which must be considered when calculating monthly returns. These are 283 pure stock splits, 153 stock dividends, 557 subscription rights and 27 reverse stock splits. The table also indicates that compared to the U.S. rights issues are more common in Germany and have to be considered carefully when calculating total rates of return.

[Table 6]

Most return differentials in the respective month of a corporate action are smaller than 5%. Only 11% of the return differentials are larger than 5%. Nevertheless, we carefully check all return differentials using our aforementioned data sources. Most return differential are caused by price differences. We find only few errors that are caused by missing or incorrect adjustment factors for pure stock splits, stock dividends and reverse stock splits. The AX time series misses three and reports five incorrect adjustment factors that cause a return differential of more than 1.0%. Thus, coverage of these corporate actions is nearly perfect. We find many errors that are caused by missing adjustments for rights issues. Datastream misses 44 (7.90%) of 557 rights issues, i.e. the AX time series provides no adjustment factor and the change in unadjusted prices, UP , is equal to the change in adjusted prices, P , ($|\Delta P - \Delta UP| < 1.0\%$) over the month when the corporate action occurred. The results of Espenlaub et al. (2009) suggest that these 44 missing adjustments are for open offers, which back then were not considered. In addition, we find 35 rights issues that are missed by the AX time series, but yield a

return differential of less than 1.0%. We also find 124 (22.26%) return differentials that are caused by different values for the subscription rights. These differences either stem from different adjustment procedures (ex-date vs. end of month) or differences in the theoretical value (or price) of the subscription rights. As argued in Section 2, these problems must be considered in studies that analyze SEOs.

4.4 Number of Shares

Presently, we find 5,811 monthly observations for 164 stocks where the number of shares between our data set and Datastream's NOSH time series differs by more than 10% for the period from 1/1990 to 10/2007. For eleven stocks (309 monthly observations) NOSH overstates the actual number of shares by a factor of at least five. For another 10 stocks (425 monthly observations) NOSH understates the number of shares by a factor of at least five. We also noticed that Datastream's NOSH time series often reports the total number of shares, thus also including shares that are not exchange listed, e.g., Deutsche Telekom AG during 11/1996 to 5/1999. There are, however, also cases where NOSH represents only the number of exchange listed shares, e.g., EnBW Energie Baden-Württemberg AG. For some firms this reporting scheme also changes over time, e.g., Pittler Maschinenfabrik AG, where NOSH includes all shares until 4/1994, but only the exchange listed shares from 4/1994 to 10/2007. In some cases such an increase/decrease in the NOSH time series might reflect changes in stocks' free float. E.g., in 2/2000 NOSH changes for Mannesman AG from 500,000k to 48,000k, thus understating the number of shares by factor of ca. ten until 8/2002. This adjustment probably reflects Vodafone's stake of 98.86% in Mannesmann. These observations are puzzling since Datastream provides data types such as NOSHC (number of shares of a company) and NOSHFF (free-float adjusted number of shares) which are supposed to take these issues into account. However, these time series are usually not available before 1/2000 (NOSHC) and 4/2002 (NOSHFF).

To obtain a clearer picture of errors in Datastream's NOSH time series, we link data regarding corporate actions to changes in the number of shares, i.e. we check whether Datastream's NOSH time series is adjusted for preceding corporate actions. In more than 90% of all cases NOSH correctly reflects the number of shares around corporate actions. Altogether we find 106 instances where the number of shares across the two data sets does not match around corporate actions. In 17 cases NOSH is updated too late, in nine cases too early, in 13 cases NOSH also represents unlisted shares, in seven cases NOSH does not distinguish between different share classes precisely (dual class firms), and in four cases NOSH is incorrectly adjusted for the corporate action. We cannot explain 37 mismatches where NOSH appears to be incorrect according to our data sources. Altogether, 87 differences are caused by errors in Datastream's NOSH time series. For the remaining 19 observations we cannot say whether Datastream or our data set provides the correct number of shares because the ex-days for in-kind contributions (Sacheinlagen) are not available to us.

By comparing an old tape (July 2009) with a recent tape (March 2012) of Datastream's NOSH time series we noticed that Datastream changed the number of shares for 48 stocks (2,971 monthly observations) that are listed in the Amtlicher Markt in Frankfurt. Some adjustments introduce errors to the time series. For example, the number of shares of Pilkington Deutschland AG was changed from 2,710k (actual number of shares) to 121k during 1/1987 and 10/2007. Similar cases occur among others for Nürnberger Beteiligungs AG (2,765k down to 27k shares) and SWARCO Traffic Holding AG (8,840k down to 547k shares). In other cases NOSH is upward adjusted. For example, for TAG Tegernsee Immobilien- und Beteiligungs-AG NOSH was changed from 9k to 56,224k shares during October 1988 and September 1998. For this stock our data set reports 900k shares when the firm was first listed in the Amtlicher Markt in Frankfurt in 10/2000, prior to this date the stock was listed in the unofficial market only. As a consequence we would allocate the stock to the portfolio of the largest stocks, D10, whereas it should be assigned to the portfolio of the smallest stocks, D01. Another example is I.G. Farbenindustrie AG where NOSH is changed from 136,000k to 1,360,000k for the period from 1/1987 to 10/2007. According to our data the number of shares is 13,600k. However, we do not consider the stocks of I.G. Farbenindustrie AG in our sample. The reasons for most changes in the NOSH time series, especially as they introduce errors remain unclear to us.

5 Empirical Results

5.1 Comparisons of Long-Term Market-Wide Rates of Return

We illustrate possible implications of using Datastream data by comparing market-wide rates of return across different data sets. For this purpose we replicate the official CDAX and the DAFOX, which are the two most frequently used stock market indices in empirical studies on the German market. Ince/Porter (2006) use this technique to document the performance of their screens, i.e. compared to raw Datastream data for the U.S. screened data produces higher correlation coefficients with broad market indices. Schmidt et al. (2011) use this technique to examine the quality of the data from Datastream. However, the limits of this technique are obvious. First, at some point minor increases/decreases in correlation coefficients become meaningless or difficult to interpret. Second, value-weight indices are usually dominated by few large stocks. Hence, having sufficient data quality for the largest stocks yields no inferences about data quality for the majority of small stocks. Annaert et al. (2011), for example, show that a value-weight index of the 20 largest stocks listed on the Brussels Stock Exchange may be considered as a good proxy for an all share index, i.e. average returns are similar and the indices are highly correlated. Third, the composition and methodology of the benchmark indices might change over time. Appendix A provides a discussion of problems that the above mentioned German indices are subject to. Based on these indices it is rather difficult to draw inferences about the quality of a data set. However, our results may have implications for the estimation of the German market risk premium.

Table 7 compares average returns on the Stehle/Hartmond-time series (incl. tax refunds), the official CDAX and the DAFOX. The official CDAX performs worst during 7/1975 and 10/2007, i.e. compared to the Stehle/Hartmond-time series the CDAX understates the performance of the German stock market by ca. 3.4 percentage points p.a. The poor performance of the CDAX is caused by different factors, among others by insufficient coverage of dividends before 1988, omitted tax refunds, and the disastrous performance of the Neuer Markt after 3/2000. Average market-wide returns from Datastream and our data set match almost perfectly for the overall period and the two subperiods, 1973 to 1990 and 1990 to 2007. This casts doubt on the high average performance of the DAFOX, which mainly results from the second sub-period, 1990 to 2007. The correlation coefficients with the official CDAX are all very close to unity.

During 1/1988 to 9/1998 the official CDAX considers all dividends and fully covers the Amtlicher Markt in Frankfurt. For this period our data set yields a rate of return of 14.59%, data from Datastream yields 13.96%. Both averages are fairly close to the average rate of return on the CDAX of 14.26%. The return on the DAFOX of 15.14%, however, overstates average returns by almost one percentage point p.a. This difference is mainly caused by a severe error in the DAFOX time series in 1998 (see Appendix A for details). Overall, the results suggest that Datastream's data quality is sufficient to estimate long-term market-wide value-weight rates of return. Nevertheless, estimates of the market risk premium based on the official CDAX, and Datastream's total return indices are too low because the tax refund of the corporate income tax paid on dividends, which German investors received between 1977 and 2000, are not considered.

[Table 7]

5.2 Descriptive Statistics and Average Returns for Size Decile Portfolios

Empirical studies for the U.S. market usually assign Amex and NASDAQ securities to portfolios using NYSE breakpoints. Fama/French (2012) argue that this sorting procedure avoids empirical results that are dominated by the plentiful small Amex and NASDAQ stocks. Following this argument, we suggest using breakpoints from the highest market segment in Germany, the Amtlicher Markt in Frankfurt. However, Datastream does not provide a history of the market segment a stock was listed. Thus, this sorting scheme is difficult to replicate. Looking at portfolios sorted by size we demonstrate how empirical results for the size effect in Germany vary with the data set and the sorting procedure. We consider three different data sources, Datastream, our data and data from the CFR. We create three different sets of size sorted portfolios using data from Datastream, DS1 to DS3. Altogether, we look at five different sets of size sorted portfolios. These portfolio sets differ with respect to the sorting procedure and coverage of the German market.

DS1) This data set includes all German equities available from Datastream. We form size decile portfolios by assigning each portfolio an equal number of stocks.

- DS2) This data set covers exactly the same stocks as DS1; portfolios are formed, however, using size breakpoints from our data set for the top segment in Frankfurt.
- DS3) This data set is a subset of DS1, it is restricted to stocks that are listed in the top segment in Frankfurt. Portfolios are also formed using size breakpoints from our data set.
- DS4) Our data set for the Amtlicher Markt in Frankfurt.
- DS5) A data set of equal-weight portfolios sorted by size which is provided by the CFR.

We consider the data from the CFR (DS5) only when we compare average portfolio returns. The data from the CFR is described in detail by Artmann et al. (2012b), basically they include primarily stocks from the Amtlicher Markt in Frankfurt and the Neuer Markt. Thus, before March 1997 (when the Neuer Markt was launched) their data set is comparable to our data set in terms of market coverage. From March 1997 onwards their data set is probably more comparable to the data set of all German equities from Datastream.

We form size portfolios at the end of June of each year t . Size is measured as the number of shares multiplied with the (unadjusted) price per share.¹¹ Equal-weight and value-weight portfolio returns are estimated for the time period from July in year t to June in year $t+1$. We look at two subperiods; the first subperiod extends from 7/1975 to 6/1990, the second from 7/1990 to 10/2007. The first subperiod starts in 1975, since we need at least two years of return data to estimate portfolios' pre-ranking rolling betas. We use rolling Dimson betas in addition to size as an independent variable in our cross-sectional Fama/MacBeth-tests. The number of stocks and the average (equal-weight) market capitalization (in real terms) per portfolio are estimated at the end of June of each year. Table 8 presents average values for these portfolio characteristics.

[Table 8]

Looking at the average number of stocks suggests that Datastream's coverage of the German stock market during 7/1975 to 6/1990 is lower than we expected. On average only ca. 42% of the stocks that are covered by the KKMDB are in Datastream (results not shown). Datastream's coverage of the Amtlicher Markt is considerably higher, on average ca. 69% of the stocks. In Section 3 we find a survivorship bias in Datastream's data for the Amtlicher Markt in Frankfurt before 1990. Comparing the number of stocks per size decile across the two data sets for the Amtlicher Markt in Frankfurt (DS3 and DS4) we also find an omission bias. Datastream's coverage of small stocks (size deciles D01 to D03) is only around 28 to 45%, whereas almost 100% of the large stocks (D08 to D10) are

¹¹ Brückner et al. (2012) estimate size by aggregating the market value of equity across all stock classes of a firm. We deviate from this procedure for two reasons. First, our results are easier to replicate when we look at stocks instead of firms. Second, we want to test whether empirical results change when we look at individual stocks instead of firms. Compared to Brückner et al. (2012) we find that the reverse size effect is statistically less pronounced for the period from 7/1990 to 10/2007.

covered by Datastream (see Table 8). As a consequence it is difficult to compare portfolios' average excess returns during the first subperiod, especially for the portfolios of small firms.

During the second subperiod from 7/1990 to 10/2007 the average number of stocks indicates that the top segment in Frankfurt represents on average only 39.34% of all German stocks. However, most stocks from the lower market segments are small and tiny stocks with average market capitalization of €4 to 107 mln. Sorting all German stocks into size deciles using breakpoints for the Amtlicher Markt in Frankfurt yields small stock portfolios with a relatively large number of stocks. On average the three portfolios of the smallest stocks represent 57% of all German stocks. However, in terms of market capitalization the top segment in Frankfurt represents on average 78% of the German exchange listed equity during 1990 and 2007. During 2000 and 2007 it represents almost 90%.

For the second subperiod the three sets of size sorted portfolios, DS2 to DS4, that are formed using breakpoints from the top segment in Frankfurt yield average excess returns that increase with size. This effect is strongest when we look at the two data sets that solely cover the Amtlicher Markt in Frankfurt (DS3 and DS4). Average return differentials between these two sets of size portfolio are relatively small and not statistically significant. Thus, the aforementioned data errors do not affect average portfolio returns. As in Brückner et al. (2012) we find some evidence of a reverse size effect in raw returns (not adjusted for risk) during the period from 7/1990 to 10/2007. The only portfolio that does not fit into this pattern is the portfolio of the smallest firms, for which we estimate an average return twice as high as for the adjacent portfolio.

Looking at the first data set of all German equities (DS1) yields a completely different return pattern. Portfolios' average excess returns decrease monotonically with size for portfolios D01 to D06, from 12.75% to -3.27%. For portfolios D06 to D10 average excess returns increase almost monotonically, from -3.27% to 5.56%. Thus, indicating a regular size effect among the smallest stocks and a reverse size effect among medium and large stocks. This pattern also emerges, but somehow less pronounced for the size portfolios from the CFR. Excess returns for the CFR size portfolios decrease from portfolio D02 (2.43%) to portfolios D04 (-3.02%) and increase from portfolio D06 (-3.54%) to D01 (6.80%). This result indicates that the size effect is not linear across all German exchange listed firms. We find a regular size effect across small firms and a reverse size effect across larger firms for the period from 7/1990 to 10/2007. Compared to the other data sets, average excess returns for the CFR portfolios are generally the lowest. The average excess returns for the portfolio of the smallest stocks, portfolio D01, in this data set of -0.02% deviates considerably from the other data sets, 3.31 to 12.75%. During the first subperiod the low returns of the CFR portfolios could be induced by coverage issues and insufficient data quality. During the second subperiod the low returns could additionally be explained by the disastrous performance of the stocks from the Neuer Markt after 2000.

5.3 Cross-Sectional Results

Looking at average returns suggests that raw portfolio excess returns are to some extent correlated with average portfolio size. We extend our analysis of the size effect in Germany using the cross-sectional test procedure of Fama/MacBeth (1973) [FM]. However, the main question we want to address is whether equity data from Datastream for the German market yield the same results on the size effect in Germany as our data set for the top segment of the Frankfurt Stock Exchange. We look at monthly excess returns of the size decile portfolios from the data sets DS1 to DS4. We use the natural logarithm of portfolios' average size and portfolios' monthly five year rolling Dimson betas (1 lag) as independent variables. We present results for equal-weight and value-weight portfolios. For equal-weight (value-weight) portfolios we estimate equal-weight (value-weight) independent variables. We look at the full time period from 7/1975 to 10/2007, but also at two subperiods. The first subperiod extends from 7/1975 to 6/1990, the second from 7/1990 to 10/2007. We choose these subperiods following our own recommendation not to use Datastream as the primary data source before 1990. Additionally, choosing these subperiods allows us to compare the results with Brückner et al. (2012), who document a strong reverse size effect for the period from 7/1990 to 10/2007. Table 9 presents the results.

For the first subperiod, the coefficient on size is mostly negative, and insignificant for the three Datastream based data sets, DS1 to DS3, when we look at value-weight portfolios, whereas the equal-weight portfolios yields mostly positive slopes on size, also not significant. Our data set for the Amtlicher Markt in Frankfurt (DS4) yields negative slopes on size for equal- and value-weight portfolios. The pricing errors are positive but usually not statistically significant across all four data sets when size is the only independent variable. The results for the slopes on size do not change as we add Dimson betas to the regression model. Thus, in our empirical setup size does not explain the cross-section of German returns during the first subperiod from 7/1975 to 6/1990. The slopes on betas are positive, but not statistically significant when we look at the two data sets that cover the top segment in Frankfurt (DS3 and DS4). The two data sets that include all German equities (DS1 and DS2) yield positive slopes on beta only for equal-weight portfolios, the slopes on beta are negative for value-weight portfolios, also not statistically significant. The pricing errors, however, usually decrease considerably and are not statistically significant anymore across all data sets as we extend the model by beta.

[Table 9]

For the second subperiod, the results differ dramatically across the four data sets. Looking at Datastream's data set of all German equities (DS1) yields negative slopes on size that are close to zero and not statistically significant. Adding beta to the model yields a slope on size that is positive for equal-weight portfolios (not statistically significant) and zero for value-weight portfolios. The slope on beta is -17.93% with a t-value of -1.53 when we look at equal-weight portfolios of all German equities

(DS1); the pricing error is positive and statistically significant at the 5% level. Sorting the same stocks using size breakpoints from our data set for the top segment yields coefficients on size that are always positive and even statistically significant at the 10% (equal-weight) and 5% (value-weight) level for the model extended by Dimson beta. The pricing errors are not statistically significant anymore. This effect is considerably more pronounced when we restrict the data set to the top segment in Frankfurt (DS3). The slope on size is now positive and statistically significant in all four variations of the test procedure. The slopes on size and the t-values are usually higher for this data set (top segment in Frankfurt) compared to the data sets of all German equities. Both data sets for the top segment (DS3 and DS4) yield slopes on size and t-values that are very similar. It is worth to mention that only if we look at value-weight portfolio from our data set we observe a positive slope on beta of ca. 1% (annualized), not statistically significant, however. All other data sets and variations in the test procedure yield negative slopes on beta after 1990.

For the overall period we observe negative slopes on size only when we look at all German equities from Datastream (DS1) and use size as the only independent variable. All other data set and variations in the test procedure yield positive coefficients on size, usually not statistically significant. Using equity data from Datastream yields generally negative slopes on Dimson beta, also not statistically significant. Portfolios from our data set yield positive slopes on beta, again not statistically significant.

We also check whether our results for the size effect are robust to grouping. Forming size deciles reduces the cross-sectional dispersion in the size variable and thus reduces efficiency for the slope on size compared to individual stocks. Looking at our dataset, DS4, we observe similar results when we look at individual stocks in our cross-sectional FM regressions and use size as the only independent variable (results not shown). For the period from 7/1990 to 10/2007 the slope on size remains positive and statistically significant, the t-values are a bit higher, though. Looking at all German equities from Datastream, DS1, now also yields positive slopes on size for the period from 7/1990 to 10/2007, not statistically insignificant, though. The slopes on size for the other periods also remain insignificant.

Overall, the results for the size effect differ considerably for the second subperiod across the four data sets. Size does not explain the cross-section of German returns when we form size decile portfolios of all German equities from Datastream and assign each portfolio an equal number of stocks. This is a sorting procedure that is frequently used in empirical studies on the German market. Following this procedure yields slopes on size that are close to zero and statistically not significant. Using data sets for the Amtlicher Markt in Frankfurt (DS3 and DS4) generally reveals a statistically significant reverse size effect during the period from 7/1990 to 10/2007. This result is in line with Brückner et al. (2012). Forming portfolios of all German equities using size breakpoints for the top segment yields similar results, but less pronounced. We claim that the difference in the results across the data sets is caused by a regular size effect within the group of the smallest stocks. Table 8 suggests a negative relationship between size and average return for the period from 7/1990 to 10/2007 within the group of

the ca. 400 smallest German stocks (deciles D01 to D06 of all German equities, DS1). Looking at the ca. 300 largest German stocks (not all are necessarily large!) listed in the top segment in Frankfurt (D02 to D10, DS4) we observe a positive relationship between size and average return.

To investigate this pattern in more detail we form two portfolios of all German equities from Datastream, DS1, i.e. we separate the stocks from the size deciles D01 to D05 (small) from those in D06 to D10 (large). Running cross-sectional FM regressions for the stocks in the small stock portfolio yields negative slopes on size that are statistically significant on the 1% level for the period from 7/1990 to 10/2007. For the large stock portfolio we obtain a positive slope on size which is also statistically significant on the 1% level. We obtain similar results when we control for betas as in Fama/French (1992). For the period from 7/1975 to 6/1990 the slope on size for small stocks is positive but not statistically significant. For the large stocks the slope is negative and statistically significant on the 10% level in Fama/French (1992) regressions. These results suggest that there might be reverse size effect among small firms for the period from 7/1975 to 6/1990. At the same time there might be a regular size effect among large firms. The results for the period from 7/1990 to 10/2007 provide evidence in favor of a regular size effect among tiny and small German stocks. Most of these stocks are not listed in the top segment of the Frankfurt stock exchange, many are OTC stocks. For the large stocks from the top segment, however, we find a strong reverse size effect. This result is in line with our results for the top market segment of the Frankfurt stock exchange. Our results suggest that the size effect in Germany differs considerably for small and large stocks and changes over time.

6 Conclusion

Ince/Porter (2006) examine the quality of equity data from Datastream for the U.S. market. We extend upon their work for the German market by comparing equity data from Datastream with equity data from our data set for German stocks listed in the top segment of the Frankfurt Stock Exchange. We document serious and systematic coverage issues in Datastream before 1990. As a consequence, we currently do not recommend using Datastream as the primary data source before 1990. Only after 1990, when serious data errors are rare, the data quality is sufficient for standard academic use. Nevertheless, we document random errors in Datastream's time series of the total return indices that are caused by incorrect adjustments for dividends and corporate actions after 1990. We additionally find a considerable number of price mismatches across the two data sets for the period from 1/1990 to 12/2000. These price differences occur because Datastream's time series for Frankfurt considers prices from other stock exchanges and/or does not consider end of month closing prices. Additional price difference occur because our copy of the KKMDB tape reports indicative prices instead of trading prices in many cases. As a consequence, we find many return differentials between our data set and Datastream. We also find important errors in Datastream's number of shares time series. Consequently, the errors in Datastream might have implications for empirical studies that include the German market.

In our empirical section, where we conduct a standard analysis of the size anomaly in Germany, we show that the use of Datastream may yield economic inferences that deviate considerable from previously reported results. Using the Fama/MacBeth (1973) test procedure, we find no reverse size effect as in Brückner et al. (2012) when we look at size decile portfolios based on data for all German equities from Datastream. The problem commences because Datastream only provides information on the market segment a stock is currently listed in. As a consequence, the plentiful, tiny stocks from the former lower and unofficial market segments dominate cross-sectional FM-regressions when we form size decile portfolios. To alleviate results that are dominated by the plentiful, tiny stocks, we advocate the use of size breakpoints from the top segment of the Frankfurt Stock Exchange. A similar procedure is advocated by Fama/French (2012) who form portfolios by sorting the universe of Amex, NASDAQ, and NYSE stocks according to NYSE breakpoints. Our results also suggest that the size effect varies considerably across different size classes. To investigate this issue, we sort all German equities into two size portfolios, small and large, and separately look at each portfolio. We find a reverse size effect among large stocks for the period from 7/1990 to 10/2007. Most of these stocks are also listed in the top segment of the Frankfurt Stock Exchange. At the same time we find a regular size effect among small German stocks. Most of these stocks are from lower market segments or OTC stocks. These and other peculiarities should be considered in empirical studies on the German market.

Furthermore, equity data from Datastream for Germany, and other developed and emerging markets, is frequently used in cross-country studies. Market coverage and data precision for Germany might be comparable to other developed markets such as, Australia, France, Japan and the UK. Data quality, however, might be considerably lower for emerging markets as for example the Czech Republic, Hungary and Turkey. This issue is also pointed out by Schmidt et al. (2011). Andrikopoulos et al. (2007), Espenlaub et al. (2009) and Rossi (2011) report several errors in Datastream data for the UK. Recently, new country specific data sets have been introduced for France and Belgium. Thus, we are awaiting further work on the quality of data from Datastream for other countries. Finally, empirical studies on the cross-section of German returns are usually based on country specific data sets. The results of these studies are difficult to replicate without access to the data or another reliable data source. Artmann et al. (2012a, 2012b) are the first to make their data set publicly available. However, currently little is known about the quality of this data set. Datastream gives us the opportunity to replicate empirical results for time periods from 1990 onward. This should be considered by future empirical studies.

Appendix A: German Stock Market Indices

In Section 5.1 compare market-wide portfolio returns with German all share stock indices to evaluate the quality of Datastream's equity data. In this section we summarize problems that these indices are subject to. The official CDAX is one of the most prominent stock indices for the German stock market. It is a (market) value-weighted index of German stocks that are listed at the Frankfurter Stock

Exchange. The index has been calculated by the Deutsche Börse AG since April 22nd, 1993 (official start date). The CDAX was recalculated for the period from December 30th, 1987 till April 22nd, 1993. Before 1988 the official CDAX is based on the FWB-Index. Therefore, the official CDAX time series is available from 1970 onwards. We identify the following problems that have to be considered:

- The CDAX currently consists of all German stocks listed in the top segment of the Frankfurt Stock Exchange, the Regulierter Markt. This segment resembles the former two official market segments in Frankfurt, the Amtlicher Markt (top segment until 2007) and the Geregelter Markt. Until September 21st, 1998 the CDAX consists only of stocks from the Amtlicher Markt. Thereafter it also represents the Geregelter Markt and the Neuer Markt, the later was closed in June 2003.¹² Thus, the composition of the CDAX changes over time. From 1998 to 2000 the number of firms represented by the CDAX nearly doubled, mainly because of IPOs in the Neuer Markt. Foreign stocks, stocks from the Freiverkehr (later Open Market) stocks that are not listed in Frankfurt are not represented by the CDAX.
- The CDAX underestimates the performance of German stocks from 1970 to 1988. This bias is caused by missing dividends and probably by a selection bias before 1988. Both issues are related to the FWB-Index, on which the official CDAX time series is based before 1988.
- The CDAX does not consider tax refunds of corporate income taxes paid on dividends, which German investors received between 1977 and 2000.¹³ As a consequence the CDAX underestimates the performance of German stocks until 2001. Historical estimates of the market risk premium based on the CDAX are, therefore, too low.
- The calculation of the CDAX was changed in 2002. Before June 24th, 2002 stocks' weights in the CDAX were quarterly estimated based on all shares outstanding. Afterwards weights are adjusted by stocks' free float.

Many studies for the German capital market as Artmann et al. (2012a, 2012b) apply the DAFOX instead of the CDAX. The DAFOX is a well documented performance index of German stocks that are listed in the Amtlicher Markt in Frankfurt.¹⁴ It is available as an equal-weight and a value-weight index. However, the DAFOX is also subject to some problems:

- From 1960 to 1974 the DAFOX does not represent all stocks of the Amtlicher Markt in Frankfurt, which induces a selection bias within this period. This problem arises because before 1974 only ca. 100 stocks are included for which daily prices are available.
- The DAFOX does not include the above mentioned tax refunds, and therefore underestimates the performance of the German stock market.
- The DAFOX is only calculated until 2004. For this reason, it is usually prolonged with the CDAX time series as in Artmann et al. (2012a, 2012b).
- The quality of the DAFOX depends on the quality of the underlying data set. The KKMDB (the underlying data set for the DAFOX) is not free of data errors. Thus, the performance of the DAFOX might be biased.
- Comparing annual returns for the value-weighted DAFOX with the official CDAX and our value-weighted return time series, we find a severe error in the DAFOX time series for the year 1998. The annual returns over the year 1998 are 15.5% for the CDAX and 18.0% for our

¹² Most stocks of the Neuer Markt were transferred to the Geregelter Markt after this market segment was closed, hence the Neuer Markt stocks are also represented by the CDAX after 2003.

¹³ See Brückner et al. (2012) for a more detailed discussion of this problem.

¹⁴ See Göppl (1995) for more details on the DAFOX.

return time series. The return on the DAFOX is, however, 32.9%, which is almost twice the CDAX return. This questions the quality of the DAFOX.

Another well documented and commonly applied proxy for the market portfolio is the Stehle/Hartmond-time series. This time series fully represents the Amtlicher Markt in Frankfurt for the time period from 1948 to 1988. For this time period it includes all financial benefits to stock holders. This time series is prolonged with the official CDAX time series from 1988 onwards. Thus, from 1988 onwards this time series is subject to the same problems as the CDAX. However, in comparison with the official CDAX this time series is fully adjusted for tax refunds.

Appendix B: Literature Overview

Data quality is generally perceived as an important aspect in empirical studies. Nevertheless, there have not been many studies that explicitly address the quality of available data sources. Most of the literature has been written for the U.S., examining the quality of the CRSP tape and COMPUSTAT. Rosenberg/Houglet (1974) are among the first to systematically examine the quality of available data sources for the U.S. market. They merge monthly price relatives from the CRSP tape with COMPUSTAT to examine differences between the two data sets. They do not consider return discrepancies between 1 and 5 percentage points as serious defects. Altogether they identify 34 “large discrepancies” (larger than 5 percentage points) between March 1962 and June 1968 which might indicate serious data errors. Only four of these discrepancies were caused by errors in the CRSP tape. The remaining errors were traced back to COMPUSTAT. Rosenberg/Houglet (1974) identify only two kind of errors, errors in recorded prices and errors in the procedures that adjusts returns for stock splits. They do not report errors related to dividends, because they only look at price relatives and not at rates of return.

Bennin (1980) updates the study of Rosenberg/Houglet (1974) for the period from 1962 to 1978. He compares rates of returns instead of price relatives, thus, considering errors in dividends as well. Compared to the previous study he reports significantly lower error rates for COMPUSTAT. This witnesses COMPUSTAT’s efforts to continuously increase data quality. However, according to Bennin (1980) COMPUSTAT’s error rate of 1 in 1,000 is still considerably higher than CRSP’s error rate of 1 in 10,000 over the years 1962 to 1978. Courtenay/Keller (1994) examine the accuracy of 718 adjustments to prices and number of shares in the CRSP tape for stock splits and stock dividends for the 1yr period from January 1st to December 21st, 1989. Most of the 142 detected discrepancies (e.g. 91 incorrect declarations, 20 ex-date differences) between CRSP and Moody's Dividend Record data, however, do not cause serious problems. Ultimately, they attest to CRSP high accuracy with respect to adjustments for stock splits and stock dividends. Shumway (1997) and Shumway/Warther (1999) find a delisting bias in the CRSP date and discuss its implication for the size effect. After correcting for the delisting bias they find no evidence for a size effect on NASDAQ before 1995.

Ince/Porter (2006) are the first to systematically examine the quality of return data from Datastream for the U.S. using reference data from CRSP. They reveal several problems in Datastream. The main problem commences because Datastream does not provide information to “easily” distinguish among different security types. For example, Datastream’s equity research lists contain non-equity securities and foreign equities. Naïve use of all securities from these lists, thus, introduces coverage issues. These coverage issues are, however, fixed by the screening techniques of Ince/Porter (2006). They demonstrate the performance of their screens comparing return data from Datastream with CRSP for size sorted portfolios. Using all securities from Datastream’s research lists yields average equal-weight returns that are too high for small firm portfolios. This problem is caused by non-equity securities in Datastream’s research lists. Excluding non-equity securities and additional screens remove major discrepancies in most size portfolios’ average returns. Ince/Porter (2006) also show that the momentum effect is not detectable using “raw” return data from Datastream. Only screening and correcting the data from Datastream yields a statistically significant momentum effect similar to the one that has been found on the basis of the CRSP tape. In addition, they find data issues which are caused by wrong dates for stock splits, missing dividends and wrong stock prices. Ince/Porter (2006) also document errors in rates of returns (based on adjusted prices) that are caused by Datastream’s restriction to report only two decimal places for prices. Thus the nature of the last problem is a technical constraint which has been solved by Datastream with the DPL-function (see footnote 3 for details).

A series of recent studies reveals several problems in Datastream’s data for the UK market. These studies use data from the London share Price Database (LSPD), Bloomberg or an own data set as reference data to analyze the quality of Datastream data. Andrikopoulos et al. (2007) document a survivorship bias in Datastream among foreign and small stocks on the defunct lists. Espenlaub et al. (2009) find that Datastream does not adjust total returns for open offers in the UK before 2/2002 which has implications for studies that analyze abnormal returns following seasoned equity offerings. Rossi (2011) finds numerous cases of incorrect price and return data in Datastream. He also argues that Datastream often incorrectly adjusts the time series of number of shares for stock splits. Overall, he concludes that, because of the serious data quality issues, Datastream should be avoided.

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Tables

Table 1: Number of securities in Datastream's constituent lists for the German market.

The table shows the total number of securities in Datastream's research lists for the German market as of March 2012 (before screening). We remove foreign securities, convertibles, subscription rights and new shares (Junge Aktien), and other non-equity securities. The last column presents the number of German equities per list (after screening). The equities from the research list are carefully consolidated and supplemented by missed securities searching Datastream for all German equities (filter: status=all, market=Germany, instrument type=equity). For each stock we obtain the time series of the total return index, RI. Only stocks for which a RI time series is available are included in our final sample of all German equities

Datastream List	Total	Foreign securities	Convertibles	Rights & new shares	Others	German equities
FGER1 & FGER2	1,470	328	9	2	2	1,129
FGERDOM	1,470	328	9	2	2	1,129
FGERKURS	1,618	409	80	3	8	1,118
DEADB1 to DEADBD6	31,306	28,553	359	173	1,365	856
All German equities	3,904	6	12	186	773	2,927
Stocks on any research list (FGER1 & FGER2, FGERDOM, FGERKURS)						1,139
Additional stocks on defunct list (DEADB1 to DEADBD6)						847
Additional stocks searching for all German equities						246
Consolidated lists (1/1973-4/2012)						2,232
Final sample (1/1973-10/2007 and RI time series available)						1,747

Table 2: List and Description of the Datastream data types that we consider in our study.

The table presents an overview of Datastream data types that we consider in our study. We list alternative data types that we do not consider for various reasons, e.g. for some alternative data types a sufficient history is not available (e.g. NOSHFF, NOSHCH), for other data types the definition does not match our needs (e.g. DPS, AF, CAI).

Data type	Description	Alternative data types
DSCD	Datastream's unique identification code for every stock (usually unique per stock, stock exchange and currency).	ISIN
UDDE	Unadjusted individual dividend, sum over all payments on the ex dividend day.	DPS, DD, UDD
DCRE	Currency of the dividend.	DCR
UP	The actual or "raw" closing price.	OP, UP.FF
P	Adjusted price, the closing price adjusted for subsequent capital actions.	P.FF
RI	Theoretical growth in value of a share holding over a specified period. Dividends are assumed to be re-invested at the closing price on the ex dividend date.	RZ, RI.FF
NOSH	Total number of ordinary shares per share class (held separately for dual class firms).	NOSHFF, NOSHCH
AX	Adjustment factor (not accumulated) for capital events such as pure stock splits and subscription rights.	AF, CAI

Table 3: Datastream's coverage of German stocks listed in the Amtlicher Markt in Frankfurt, 1973-2007.

The table compares the number of stocks in our data set with the number of stocks in Datastream for the top segment in Frankfurt during the period from 1/1973 to 10/2007. The overall period is subdivided into two subperiods, 1/1973-12/1989, and 1/1990-10/2007. The table displays the number of stocks available from Datastream's research lists (excluding and including the defunct lists) and all German equities (filter: status=all, market=Germany, instrument type=equity). The bottom part of the table shows for each subperiod the number of stocks for which a total return time series, RI, is available (at least for one month) and those for which a complete total return time series is available.

Market coverage	1/1973-12/1989		1/1990-10/2007	
	Obs.	(in %)	Obs.	(in %)
Number of stocks in our data set	338		576	
Number of stocks in Datastream's:				
- Research lists, excl. defunct list	104	(30.77)	299	(51.91)
- Research lists, incl. defunct lists	297	(87.87)	571	(99.13)
- List of all German equities	300	(88.76)	575	(99.83)
Return data coverage of Datastream				
Returns time series (RI) available	289	(85.50)	574	(99.65)
Full return time series (RI) available	207*	(61.42)	564**	(97.92)

* Incomplete RI time series; the RI time series start on average 132 month too late. **Incomplete RI time series; the RI time series start on average 26 month too late, this is largely caused by two firms, Otto Stumpf AG (168 month missing) and Escom AG (56 month missing).

Table 4: Monthly return differentials between Datastream and our data set for German stocks listed in the Amtlicher Markt in Frankfurt, 1/1973-10/2007.

The first row (number of returns) presents the number of monthly returns according to our data set for the Amtlicher Markt in Frankfurt. The second row presents the number of monthly returns that we could merge with Datastream. Matched returns illustrates the number of returns across the two data set that match, i.e. the number of merged returns where the return differential is less than 1%. The bottom part of the table presents the number of return differentials that are larger than 1, 5, 10 and 15%. We subdivide the overall period (1/1973-10/2007) into three periods to illustrate how the number of return differentials varies through time.

	1/1973-10/2007		1/1973-12/1989		1/1990-12/2000		1/2001-10/2007	
	Obs.	(in %)	Obs.	(in %)	Obs.	(in %)	Obs.	(in %)
Number of returns	119,133		45,398		45,151		28,584	
Merged returns	103,992	87.29	30,819	67.89	44,690	98.98	28,480	99.64
Matched returns	84,649	81.40	27,239	88.38	30,200	67.58	27,207	95.53
Return differentials (RD)								
RD >= 1%	19,343	18.60	3,580	11.62	14,490	32.42	1,273	4.47
RD >= 5%	2,701	2.60	295	0.96	1,958	4.38	448	1.57
RD >=10%	756	0.73	81	0.26	477	1.07	198	0.70
RD >=15%	333	0.32	36	0.12	203	0.45	94	0.33

Table 5: Average number of dividends, dividend yields and return differentials in dividend payout months of German stocks traded in the Amtlicher Markt in Frankfurt, 1/1973-10/2007.

The table compares the number of dividend payments, dividend yields and return differentials in dividend payout months between our data set and data from Datastream for German stocks listed in the Amtlicher Markt in Frankfurt. We look at different time periods during 1973 and 2007 to show how Datastream's coverage of dividend payments improves over time. For our data set we show the average number of dividend payments and dividend yields (D/UP) per year. D denotes the dividend (incl. bonuses) and UP denotes Datastream's unadjusted price on the cum dividend day. Looking at dividend data from Datastream we present the average number of dividends per year in the UDDE time series for German stocks listed in the Amtlicher Markt in Frankfurt and the fraction of dividend payments according to our data set (in % of AM). We present average dividend yields per year which are based on data from Datastream using two different dividend yield measures, UDDE/UP and dRI-dP. UDDE is the dividend from Datastream. The latter measure is based on the monthly differences between the change in the total return index (dRI) and the change in the adjusted prices (dP). This measure considers only observations where dRI is larger than dP. The two measures consider only stocks for which Datastream's UDDE time series is not void. The last two columns show average return differentials per year between the two data sets. Return differentials are estimated as the difference between the monthly rate of return from our data set, R, and the change in the total return index from Datastream, dRI. To calculate this measure we consider only months for which (i) our data set reports a dividend payout and (ii) we can calculate a monthly rate of return using Datastream's RI time series over the month of the dividend payment.

Period	Amtlicher Markt		Datastream				Return Differentials	
	Number of dividends	D/UP (in %)	Number of dividends	in % of AM	UDDE/UP (in %)	dRI-dP (in %)	Number of RDs	R-dRI (in %)
1/1973-12/1980	183.4	3.39	-	-	-	-	115.75	3.22
1/1981-12/1987	189.1	2.79	21.9	11.56	2.89	2.89	133.14	2.36
1/1988-12/1988	243.0	2.63	130.0	53.50	2.50	2.36	177.00	1.33
1/1989-12/1989	250.0	2.19	162.0	64.80	2.19	2.63	246.00	0.74
1/1990-12/1990	269.0	1.99	262.0	97.40	1.95	2.69	264.00	0.10
1/1991-12/1999	283.1	2.58	278.9	98.51	2.53	2.69	282.78	0.03
1/2000-10/2007	220.3	3.25	218.6	99.26	3.24	3.04	219.88	0.19

Table 6: Relevant capital adjustments, 1/1990 – 10/2007.

The table presents capital actions (pure stock splits, stock dividends, subscription rights, reverse stock splits, and others) for all German stocks listed in the Amtlicher Markt in Frankfurt during the period from January 1990 to October 2007. Obs. represents the number of observation for each group of capital actions. For each observation we compare monthly rates of return over the month of the capital action using data from Datastream, RI, and our data set. Panel A provides the absolute and relative number of return differentials within each event group that are larger than 1.0, 5.0, and 10.0%. The last column in Panel A states the average return differential over all events within an event group. Panel B provides information on what causes return differentials larger than 1.0%. Possible sources are different prices, different adjustment factors, or missing adjustment factors in Datastream. Return differentials that are not solely explained by different stock prices or adjustment factors are summarized in the last column of Panel B.

Panel A: Number of absolut return differentials higher than 1.0%, 5.0%, and 10.0%.

Capital action	Obs.	$ RD \geq 1\%$		$ RD \geq 5\%$		$ RD \geq 10\%$		Avg. RD
Pure stock splits	283	102	36%	30	11%	11	4%	0.54%
Stock dividends	153	38	25%	11	7%	1	1%	0.15%
Rights issue	557	221	40%	57	10%	20	4%	-0.23%
Rev. stock splits	27	14	52%	13	48%	9	33%	-5.42%
Others	3	-	-	-	-	-	-	-
Total	1028	375	36%	111	11%	41	4%	

Panel B: Absolut return differentials higher than 1.0% and their cause.

	Obs.	Different Price		Different Adj. Factor		Missing Adj. Factor		Other	
Pure stock splits	102	82	80%	2	2%	1	1%	17	17%
Stock dividends	38	27	71%	2	5%	1	3%	8	21%
Rights issue	221	24	11%	124	56%	44	20%	29	13%
Rev. stock splits	14	2	14%	1	7%	1	7%	10	71%

Table 7: Comparisons of average market-wide rates of return, 1/1975-10/2007.

The table compares average annualized rates of returns (in %) of the most frequently used German all share indices (Stehle/Hartmond-time series, the official CDAX and the DAFOX) and market-wide value-weight portfolios formed using data from Datastream and our data set. The DAFOX is expanded for the period from 1/2005 to 10/2007 with the official CDAX. We look at two data sets from Datastream, the first considers all German equities (TDS, Germany), the second only those equities that are listed in the Amtlicher Markt in Frankfurt (TDS, Amtlicher Markt). Our data set also considers only stocks listed in the top segment of the Frankfurt stock exchange, the Amtlicher Markt. We subdivide the overall period from 1/1975 to 10/2007 into two subperiods (7/1975-6/1990 and 7/1990-10/2007). In addition, we consider the period from 1/1988 to 9/1998 when the CDAX considers only stocks from the Amtlicher Markt in Frankfurt and includes all dividends.

	7/1975-10/2007	7/1975-6/1990	7/1990-10/2007	1/1988-9/1998
Data set	Average annualized rate of return (in %)			
Stehle/Hartmond	12.21	14.63	10.12	15.32
Official CDAX	10.31	11.23	9.52	14.26
DAFOX	12.00	13.15	10.99	15.14
TDS, Germany	11.06	13.05	9.33	13.87
TDS, Amtlicher Markt	11.40	12.98	10.03	13.96
Our data, Amtlicher Markt	11.58	13.11	10.25	14.59
	Correlation with CDAX			
DAFOX	0.9775	0.9911	0.9812	0.9888
TDS, Amtlicher Markt	0.9829	0.9822	0.9933	0.9944
Our data, Amtlicher Markt	0.9826	0.9908	0.9882	0.9962

Table 8: Descriptive statistics and equal-weight excess rates of return for size sorted decile portfolios, 7/1975-10/2007.

The table presents the average number of stocks, average market capitalization in prices of 2007 and average excess returns (portfolios' return less the risk-free rate, SU0104, annualized, in percent) for different size sorted portfolios for the German market. We look at five different data sets, DS1 to DS5. DS1) Datastream data for all German equities. DS2) Datastream data for all German equities sorted using size breakpoints. DS3) Datastream data for German stocks listed in the Amtlicher Markt in Frankfurt sorted using size breakpoints. DS4) Our data set for the Amtlicher Markt in Frankfurt. DS5) Data from the CFR (only average returns available to us). We form size portfolios at the end of June of every year t . Size is measured by a stock's market value of equity at the end of June in year t (number of shares multiplied with the stock price). We assign the same number of stocks to each size decile for data set DS1 and DS4. The number of stocks per size portfolio differ for data sets DS2 and DS3 due to the use of size breakpoints. The size breakpoints are estimated from our data set for the Amtlicher Markt in Frankfurt. Equal-weight portfolio returns are calculated for the period from July in year t to June in year $t+1$. The data from the CFR is described in detail by Artmann et al. (2012b). We present results for two subperiods, 7/1975-6/1990 and 7/1990-10/2007.

Data set	7/1975-6/1990										7/1990-10/2007									
	D01 (Small)	D02	D03	D04	D05	D06	D07	D08	D09	D10 (Large)	D01 (Small)	D02	D03	D04	D05	D06	D07	D08	D09	D10 (Large)
	Average number of stocks										Average number of stocks									
DS1) TDS, all German stocks	30.9	26.4	26.4	26.4	26.4	26.4	26.4	26.4	26.4	26.4	88.8	84.8	84.7	84.8	84.8	84.8	84.8	84.8	84.8	84.8
DS2) TDS, all German stocks, breakpoints	38.9	26.4	24.9	22.3	28.9	26.8	23.3	24.1	25.4	26.3	282.8	119.7	84.1	68.4	61.2	57.7	54.3	43.7	41.4	37.9
DS3) TDS, top segment, breakpoints	7.9	10.2	10.3	13.9	17.6	16.6	18.8	21.6	21.2	23.3	33.6	35.7	34.0	32.1	32.1	32.4	33.8	34.1	34.1	34.0
DS4) Our data, top segment	26.5	22.4	22.4	22.3	22.4	22.4	22.4	22.4	22.4	22.4	38.1	33.6	33.6	33.6	33.6	33.6	33.6	33.6	33.6	33.6
	Average market capitalization (in mln. Euro, real)										Average market capitalization (in mln. Euro, real)									
DS1) TDS, all German stocks	17	48	88	127	181	273	435	639	1,257	5,535	4	12	24	41	66	107	183	344	823	9,170
DS2) TDS, all German stocks, breakpoints	12	37	70	113	166	250	408	612	1,115	4,972	15	51	89	145	228	366	627	1,187	2,748	16,186
DS3) TDS, top segment, breakpoints	16	37	72	114	165	255	407	613	1,117	4,493	21	52	90	147	230	368	636	1,196	2,767	16,535
DS4) Our data, top segment	14	38	71	113	166	258	410	613	1,156	5,219	20	53	90	148	231	370	642	1,202	2,800	17,600
	Average equal-weight excess return (annualized, in %)										Average equal-weight excess return (annualized, in %)									
DS1) TDS, all German stocks	6.56	7.49	6.93	8.00	7.42	8.35	8.26	6.33	5.53	7.49	12.75	2.92	2.54	1.39	-1.02	-3.27	0.83	-0.23	3.75	5.56
DS2) TDS, all German stocks, breakpoints	3.82	8.23	7.06	6.96	9.08	5.74	10.93	6.28	5.77	7.05	6.19	0.76	-2.31	-0.82	2.83	0.52	2.70	2.09	6.48	5.03
DS3) TDS, top segment, breakpoints	7.76	6.09	10.24	9.45	7.28	6.14	10.52	6.26	5.77	7.21	3.45	0.30	-1.55	0.80	4.69	3.81	4.34	4.89	6.42	7.03
DS4) Our data, top segment	10.15	7.07	8.15	6.91	7.46	6.22	9.06	5.45	6.27	7.24	3.39	-0.79	-0.79	1.19	5.28	4.03	5.13	4.82	6.11	7.65
DS5) CFR	8.60	5.61	6.80	6.04	6.61	7.25	8.02	6.81	6.96	5.22	-0.02	2.43	-2.65	-3.02	-2.34	-3.54	-0.90	-1.41	4.60	6.80

Table 9: Results for cross-sectional Fama/MacBeth regressions for one-dimensional sorted size decile portfolios, 7/1975-10/2007.

The table presents average intercepts (annualized alphas in %) and slopes on portfolio characteristics (annualized in %) from monthly cross-sectional Fama/MacBeth regressions of excess returns on size and beta. We present results for equal-weight and value-weight size decile portfolios. We use monthly excess rates of return as dependent variable in the regressions. The independent variables are determined by a portfolio's average size and 5-yr rolling beta. Monthly Dimson betas are estimated regressing 60 (at least 24) monthly portfolio excess returns on the market excess return and one-month lagged market excess return. We look at the overall period from 7/1975 to 10/2007, as well as two subperiods, 7/1975-6/1990 and 7/1990-10/2007. The t-values (Newey/West adjusted) for the intercepts and slopes are presented in parentheses. Values that are statistically significant different from zero at the five percent level are highlighted.

7/1975-10/2007						7/1975-6/1990						7/1990-10/2007					
Equal-weight			Value-weight			Equal-weight			Value-weight			Equal-weight			Value-weight		
Alpha	Beta	Size	Alpha	Beta	Size	Alpha	Beta	Size	Alpha	Beta	Size	Alpha	Beta	Size	Alpha	Beta	Size
1) Datastream data for all German equities																	
5.99		-0.23	6.16		-0.17	7.57		0.03	8.94		-0.27	4.62		-0.45	3.74		-0.09
(1.71)		(-0.70)	(1.64)		(-0.48)	(1.53)		(0.06)	(1.91)		(-0.57)	(0.92)		(-1.02)	(0.64)		(-0.17)
8.52	-7.98	0.38	6.54	-1.40	0.04	4.10	3.53	0.29	8.55	-0.67	0.08	12.34	-17.93	0.46	4.80	-2.03	0.00
(1.81)	(-1.00)	(0.84)	(1.28)	(-0.18)	(0.09)	(0.50)	(0.37)	(0.55)	(1.03)	(-0.08)	(0.17)	(2.19)	(-1.53)	(0.68)	(0.76)	(-0.17)	(0.00)
2) Datastream data for all German equities, sorted using breakpoints from our data set for the Amtlicher Markt in Frankfurt																	
2.53		0.41	3.20		0.36	6.54		0.24	8.03		-0.08	-0.94		0.56	-0.97		0.73
(0.68)		(1.22)	(0.86)		(1.04)	(1.27)		(0.52)	(1.64)		(-0.17)	(-0.17)		(1.26)	(-0.18)		(1.55)
1.41	-1.51	0.75	7.03	-6.10	0.58	-0.09	6.50	0.47	11.70	-4.05	-0.01	2.72	-8.45	0.99	2.98	-7.87	1.09
(0.34)	(-0.28)	(1.92)	(1.60)	(-1.19)	(1.48)	(-0.01)	(0.79)	(0.93)	(1.52)	(-0.51)	(-0.02)	(0.64)	(-1.18)	(1.71)	(0.65)	(-1.16)	(1.99)
3) Datastream data for the Amtlicher Markt in Frankfurt, sorted using breakpoints from our data set for the Amtlicher Markt in Frankfurt																	
2.63		0.50	4.04		0.26	9.40		-0.24	11.21		-0.59	-3.23		1.14	-2.17		0.99
(0.69)		(1.31)	(1.06)		(0.63)	(1.72)		(-0.43)	(2.09)		(-0.99)	(-0.60)		(2.52)	(-0.39)		(1.86)
-0.71	-0.20	0.90	-0.56	1.19	0.65	1.65	5.19	0.27	1.99	7.00	-0.15	-2.76	-4.86	1.44	-2.76	-3.83	1.35
(-0.18)	(-0.04)	(2.06)	(-0.14)	(0.23)	(1.45)	(0.27)	(0.95)	(0.45)	(0.31)	(1.14)	(-0.24)	(-0.56)	(-0.60)	(2.37)	(-0.49)	(-0.47)	(2.06)
4) Our data for the Amtliche Markt in Frankfurt																	
2.77		0.45	2.49		0.50	9.88		-0.41	8.06		-0.13	-3.39		1.20	-2.33		1.04
(0.76)		(1.24)	(0.69)		(1.31)	(2.28)		(-0.80)	(1.86)		(-0.24)	(-0.61)		(2.52)	(-0.41)		(1.98)
-0.79	1.32	0.65	-2.13	3.18	0.65	1.50	9.76	-0.53	2.10	5.65	-0.04	-2.77	-5.99	1.67	-5.80	1.05	1.24
(-0.20)	(0.23)	(1.50)	(-0.52)	(0.55)	(1.48)	(0.23)	(1.19)	(-1.01)	(0.33)	(0.68)	(-0.07)	(-0.57)	(-0.77)	(2.68)	(-1.12)	(0.12)	(1.84)